

Research Question 1

GPA & Extension Requests

Question

Will students with lower starting GPAs more likely request extensions than students with higher starting GPAs?

Hypothesis

Students with lower starting GPAs are more likely to request extensions than students with higher starting GPAs.

PUT THIS ON THE PAPER!!!

$$p = 0.0001$$

$$\rho = -0.38$$

This means there is a significant moderate negative correlation between GPA and Extensions Requested

Raw Values Summary

OLD DATA

1. GPA Group 2.0 – 2.49 (with a count of 3) has 1.3333 average extensions used
2. GPA Group 2.5 – 2.99 (with a count of 8) has 1.875 average extensions used
3. GPA Group 3.0 – 3.49 (with a count of 17) has 1.6471 average extensions used
4. GPA Group 3.5 – 4.0 (with a count of 41) has 1.122 average extensions used
5. GPA Group 4.0 + (with a count of 8) has 0.625 average extensions used

NEW DATA

1. GPA Group 2.0 – 2.49 (with a count of 5) has 1.6000 average extensions used
2. GPA Group 2.5 – 2.99 (with a count of 12) has 1.5000 average extensions used
3. GPA Group 3.0 – 3.49 (with a count of 24) has 1.7500 average extensions used
4. GPA Group 3.5 – 4.0 (with a count of 42) has 1.1905 average extensions used
5. GPA Group 4.0 + (with a count of 8) has 0.500 average extensions used

	GPA_Group	n	Mean_Extensions	SE
1	2.0 - 2.49	5	1.6000	0.4000000
2	2.5 - 2.99	12	1.5000	0.1946247
3	3.0 - 3.49	24	1.7500	0.1085143
4	3.5 - 4.0	42	1.1905	0.1285951
5	4.0+	8	0.5000	0.1889822

- SE is Standard Error, the sample size and the standard deviation (how wildly different the students answered) both play a role in the error.

Statistical Note

Test used: Spearman Correlation

Result: OLD: $\rho = -0.43, p < 0.0001$

NEW RESULT

$$\rho = -0.38, p = 0.00018 < 0.001$$

Spearman's rank correlation rho

```
data: Cleaned_RQ01_Data$GPA_Rank and Cleaned_RQ01_Data$num_exts_Requested
S = 173702, p-value = 0.0001775
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.3831952
```

Conclusion

OLD CONCLUSION: Using a Spearman Correlation with a rho value of -0.43 and a P-value of < 0.0001 . This means there is a moderate negative correlation; as the GPA increases, the number of extensions requested decreases. The bar graph shows a clear downward trend. The reason there are no entries for the “1.5 - 1.99” rank is due to no IRB approved students answering this.

NEW CONCLUSION: Using a Spearman Correlation with a rho value of -0.38 and a P-value of < 0.001 . This means there is a moderate negative correlation; as the GPA increases, the number of extensions requested decreases. The bar graph shows a clear downward trend. The reason there are no entries for the “1.5 - 1.99” rank is due to no IRB approved students answering this.